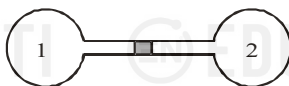


Subjective Type

$$\therefore \beta = 8$$

1. Two spherical flasks having volume $V_0 = 1.0$ L each containing air are connected by a tube of diameter $d = 6$ mm and length $l = 1$ m. A small droplet of mercury contained in the tube is at its middle at 0°C . By what distance do the mercury droplet move if the flask 1 is heated by 2°C while flask 2 is cooled by 2°C . Ignore any expansion of flask wall.



Solution

$$V_0 = 1\text{L} = 10^{-3}\text{ m}^3, T_0 = 273\text{K}, r = 3\text{mm} = 3 \times 10^{-3}\text{ m}$$

For both flasks, $PV = nRT$

$$\Rightarrow \frac{\Delta P_1}{P_0} + \frac{\Delta V_1}{V_0} = \frac{\Delta T_1}{T_0} \quad \text{and} \quad \frac{\Delta P_2}{P_0} + \frac{\Delta V_2}{V_0} = \frac{\Delta T_2}{T_0}$$

Here, $\Delta V_1 + \Delta V_2 = 0$ and $\Delta T_1 + \Delta T_2 = 2 - 2 = 0$.

$$\Rightarrow \Delta P_1 + \Delta P_2 = 0.$$

As the pressure in flask 1 is always equal to the pressure in flask 2, we have

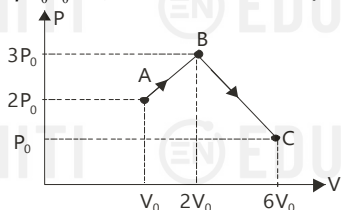
$$\Delta P_1 = \Delta P_2 = 0. \quad \Rightarrow \Delta V_1 = V_0 \Delta T_1 / T_0$$

If the mercury droplet shifts by Δx , then

$$\Delta x = \frac{\Delta V}{\pi r^2} = \frac{V_0 \Delta T / T_0}{\pi r^2} = \frac{10^{-3} \times 2 / 273}{\pi \times (3 \times 10^{-3})^2} = 0.259\text{m}$$

$$= \boxed{25.9\text{cm}}$$

2. One mole of monatomic ideal gas undergoes a process ABC as shown in figure. If the maximum temperature of the gas during the process ABC is $\beta P_0 V_0 / R$, then the value of β is



Solution For maximum temperature, product of P and V must be maximum. From $A \rightarrow B$, maximum product is at B i.e. $6P_0 V_0$.

For $B \rightarrow C$, equation of P vs V is

$$P = -\frac{P_0}{2V_0}(V - 2V_0) + 3P_0 = -\frac{P_0}{2V_0}V + 4P_0$$

$$\text{So, product } PV = -\frac{P_0}{2V_0}V^2 + 4P_0V$$

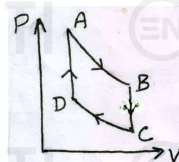
$$\text{For maximum, } \frac{d(PV)}{dV} = 0$$

$$\Rightarrow -\frac{P_0}{2V_0} \times 2V + 4P_0 = 0 \quad \Rightarrow V = 4V_0$$

$$\text{When } V = 4V_0, P = -\frac{P_0}{2V_0} \times 4V_0 + 4P_0 = 2P_0$$

$$\text{Hence, } T_{\max} = \frac{PV}{R} = \frac{2P_0 \times 4V_0}{R} = \frac{8P_0 V_0}{R}$$

3. On mole of a monatomic ideal gas is taken through the cycle shown in figure.



Processes $A \rightarrow B$ and $C \rightarrow D$ are adiabatic and

Processes $B \rightarrow C$ and $D \rightarrow A$ are isochoric.

Given that $T_A = 1000$ K, $P_B = (2/3) P_A$ and $P_C = (1/3) P_A$.

Calculate the following quantities:

(a) The work done by the gas in the process $A \rightarrow B$

(b) The heat lost by the gas in the process $B \rightarrow C$

(c) The temperature T_D .

(Take $(2/3)^{2/5} = 0.85$)

Solution As process AB is adiabatic,

$$P_A^{(1-\gamma)/\gamma} T_A = P_B^{(1-\gamma)/\gamma} T_B$$

$$\Rightarrow T_B = \left(\frac{P_B}{P_A} \right)^{(\gamma-1)/\gamma} T_A \quad \text{where} \quad \frac{\gamma-1}{\gamma} = \frac{5/3-1}{5/3} = \frac{2}{5}$$

$$\Rightarrow T_B = \left(\frac{2}{3} \right)^{2/5} \times 1000 = 0.85 \times 1000 = 850\text{ K}$$

$$(a) \Delta W_{AB} = \frac{nR(T_B - T_A)}{1-\gamma} = \frac{1 \times 8.31 \times (850 - 1000)}{1-5/3} = \boxed{1870\text{J}}$$

$$(b) \text{ It is given that } P_A = \frac{3}{2} P_B = 3P_C \quad \Rightarrow P_B = 2P_C$$

As process BC is isochoric, $P \propto T$

$$\Rightarrow \frac{T_C}{T_B} = \frac{P_C}{P_B} = \frac{1}{2} \quad \Rightarrow T_C = \frac{1}{2} \times 850 = 425\text{ K}$$

$$\Delta Q_{BC} = nC_V(T_C - T_B) = 1 \times \frac{3}{2} \times 8.31 \times (425 - 850) = -5298\text{ J}$$

$\therefore$ Heat lost by the gas in process BC is $\boxed{5298\text{J}}$

(c) As processes AB and CD are adiabatic,

$$T_A V_A^{\gamma-1} = T_B V_B^{\gamma-1} \quad \text{and} \quad T_C V_C^{\gamma-1} = T_D V_D^{\gamma-1}$$

$$\Rightarrow \frac{T_A}{T_B} = \left(\frac{V_B}{V_A} \right)^{\gamma-1} = \left(\frac{V_C}{V_D} \right)^{\gamma-1} = \frac{T_D}{T_C}$$

$$\therefore T_D = \frac{T_A}{T_B} \times T_C = \frac{1000}{850} \times 425 = \boxed{500\text{K}}$$



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4. A massless piston divides a closed, thermally insulated cylinder into two equal parts. One part contains 28g of nitrogen. At this temperature, one-third of molecules are dissociated into atoms and the other part is evacuated. The piston is released and the gas fills the whole volume of the cylinder at temperature T_0 . Then, the piston is slowly displaced back to its initial position. Calculate the increase in internal energy of the gas. Neglect further dissociation of molecules during the motion of the piston.

Solution. Let the volume of each part be V_0 .

Initially, number of moles of N_2 are $n = m/M = 28/28 = 1$

After one-third molecules are dissociated into atoms, the number of moles of monatomic N are $n_1 = 2 \times 1/3 = 2/3$

and number of moles of diatomic N_2 are $n_2 = 1 - 1/3 = 2/3$.

Degree of freedom of monatomic and diatomic nitrogen respectively are $f_1 = 3$ and $f_2 = 5$.

$$\Rightarrow f_{\text{mix}} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} = \frac{2/3 \times 3 + 2/3 \times 5}{2/3 + 2/3} = 4$$

$$\Rightarrow \gamma = 1 + \frac{2}{f_{\text{mix}}} = 1 + \frac{2}{4} = 1.5 \quad \text{and} \quad C_v = \frac{f_{\text{mix}}}{2} R = 2R$$

Before the piston is brought back to its initial position,

$$T_1 = T_0 \quad \text{and} \quad V_1 = 2V_0$$

After the piston is brought back to its initial position, $V_2 = V_0$ and let the final temperature be T_2 . As the process is adiabatic,

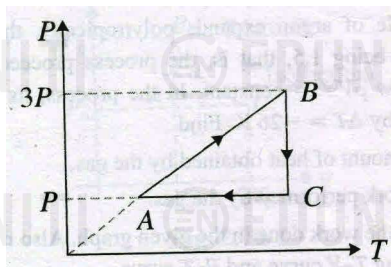
$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\Rightarrow T_2 = T_1 (V_1/V_2)^{\gamma-1} = T_0 (2V_0/V_0)^{1.5-1} = \sqrt{2} T_0$$

$$\therefore \Delta U = n C_v \Delta T = (n_1 + n_2) \times 2R \times (T_2 - T_1)$$

$$= \left(\frac{2}{3} + \frac{2}{3} \right) \times 2R \times (\sqrt{2} T_0 - T_0) = \left[\frac{8}{3} R T_0 (\sqrt{2} - 1) \right]$$

5. A fixed mass of oxygen gas performs a cyclic process $ABCA$ as shown. Find the efficiency of the process.



Solution. Oxygen is a diatomic gas. So,

$$C_v = 5R/2 \quad \text{and} \quad C_p = 7R/2$$

$A \rightarrow B$ is an isochoric process. So,

$$\frac{P_A}{T_A} = \frac{P_B}{T_B} \quad \Rightarrow T_B = 3T_A$$

$$\Delta Q_{AB} = n C_v (T_B - T_A) = n \times \frac{5}{2} R \times (3T_A - T_A) = 5nRT_A$$

$B \rightarrow C$ is an isothermal process. So, $T_C = T_B = 3T_A$

$$\Delta Q_{BC} = \Delta W_{BC} = nRT_B \ln(P_B/P_C) = 3nRT_A \ln 3$$

$C \rightarrow A$ is an isobaric process.

$$\Delta Q_{CA} = n C_p (T_A - T_C) = n \times \frac{7}{2} R \times (T_A - 3T_A) = -7nRT_A$$

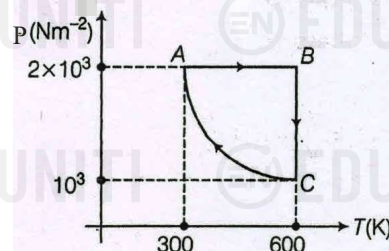
Net work done is

$$\Delta W_{\text{net}} = \Delta Q_{\text{net}} - \Delta U_{\text{net}} = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA} - 0$$

$$= 5nRT_A + 3nRT_A \ln 3 - 7nRT_A = nRT_A \ln 3 - 2nRT_A$$

$$\therefore \eta = \frac{\Delta W_{\text{net}}}{\Delta Q_{\text{in}}} = \frac{nRT_A \ln 3 - 2nRT_A}{5nRT_A + 3nRT_A \ln 3} = \frac{3 \ln 3 - 2}{5 + 3 \ln 3}$$

6. Two moles of a monatomic ideal gas is taken through a cyclic process as shown.



Process CA follows $PT = \text{constant}$. Find the cycle efficiency

Solution Process $A \rightarrow B$ is isobaric.

$$\Rightarrow \Delta Q_{AB} = n C_p (T_B - T_A) = 2 \times \frac{5R}{2} \times (600 - 300) = 1500R$$

Process $B \rightarrow C$ is isothermal.

$$\Rightarrow \Delta Q_{BC} = \Delta W_{BC} = nRT_B \ln \frac{P_B}{P_C} = 2 \times R \times 600 \ln \left(\frac{2 \times 10^3}{10^3} \right)$$

$$= 1200R \ln 2$$

In process $C \rightarrow A$, $PT = \text{const.} \Rightarrow PV^{1/2} = \text{const.}$

It is a polytropic process with $\alpha = 1/2$

$$\Rightarrow C = C_v + \frac{R}{1-\alpha} = \frac{3R}{2} + \frac{R}{1-1/2} = \frac{7R}{2}$$

$$\Rightarrow \Delta Q_{CA} = nC(T_A - T_C) = 2 \times \frac{7R}{2} \times (300 - 600) = -2100R$$

Net work done in complete cycle,

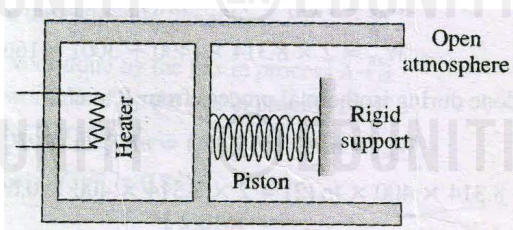
$$\Delta W_{\text{net}} = \Delta Q_{\text{net}} - \Delta U_{\text{net}} = (\Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA}) - 0$$

$$= 1500R + 1200R \ln 2 - 2100R = 1200R \ln 2 - 600R$$

$$\therefore \eta = \frac{\Delta W_{\text{net}}}{\Delta Q_{\text{in}}} = \frac{1200R \ln 2 - 600R}{1500R + 1200R \ln 2} = \frac{12 \ln 2 - 6}{12 \ln 2 + 15}$$

$$= \frac{12 \times 0.693 - 6}{12 \times 0.693 + 15} = 0.0993 = 9.93\%$$

7. An ideal monatomic gas is confined in a cylinder by a massless and frictionless spring loaded piston of cross sectional area $8 \times 10^{-3} \text{ m}^2$. Initially, the gas is at 300 K and occupies a volume of $2.4 \times 10^{-3} \text{ m}^3$ and the spring is in its relaxed state. Now the gas is heated by a small electric heater until the piston moves out slowly by 0.1 m . Calculate the final temperature of the gas, the change in internal energy of the gas, the work done by the gas and the heat supplied by the heater. The cylinder and the piston are thermally insulated. The force constant of the spring is 8000 N/m and atmospheric pressure is $P_0 = 1 \times 10^5 \text{ N/m}^2$.



Solution. The displacement of piston is $x_0 = 0.1 \text{ m}$

Initially, $P_1 = P_0 = 10^5 \text{ N/m}^2$, $V_1 = 2.4 \times 10^{-3} \text{ m}^3$, $T_1 = 300 \text{ K}$

Finally, $P_2 = P_0 + \frac{kx_0}{A} = 10^5 + \frac{8000 \times 0.1}{8 \times 10^{-3}} = 2 \times 10^5 \text{ N/m}^2$

$V_2 = V_1 + Ax_0 = 2.4 \times 10^{-3} + 8 \times 10^{-3} \times 0.1 = 3.2 \times 10^{-3} \text{ m}^3$

Using gas equation $P_1 V_1 / T_1 = P_2 V_2 / T_2$, the final temperature is

$$T_2 = \left(\frac{P_2}{P_1} \right) \left(\frac{V_2}{V_1} \right) T_1 = 2 \times \frac{3.2}{2.4} \times 300 = \boxed{800 \text{ K}}$$

The change in internal energy of the gas is

$$\Delta U = n C_V \Delta T = \frac{P_1 V_1}{RT_1} \times \frac{3}{2} R (T_2 - T_1) = \frac{3}{2} P_1 V_1 \left(\frac{T_2}{T_1} - 1 \right)$$

$$= \frac{3}{2} \times 10^5 \times 2.4 \times 10^{-3} \times \left(\frac{800}{300} - 1 \right) = \boxed{600 \text{ J}}$$

The work done by the gas is

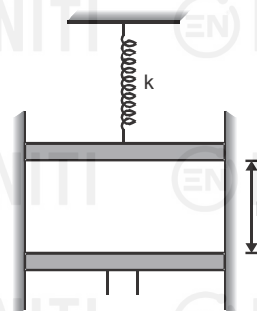
$$\Delta W = \int P dV = \int_0^{x_0} \left(P_0 + \frac{kx}{A} \right) A dx = P_0 A x_0 + \frac{1}{2} k x_0^2$$

$$= 10^5 \times 8 \times 10^{-3} \times 0.1 + \frac{1}{2} \times 8000 \times 0.1^2 = \boxed{120 \text{ J}}$$

The heat supplied by the heater is

$$\Delta Q = \Delta U + \Delta W = 600 + 120 = \boxed{720 \text{ J}}$$

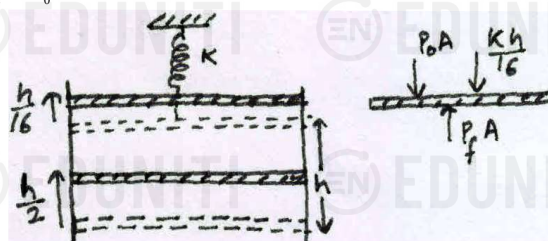
8. An ideal gas at NTP is enclosed in a adiabatic vertical cylinder having area of cross section $A = 27 \text{ cm}^2$, between two light movable adiabatic pistons as shown in the figure. Spring with force constant $k = 3700 \text{ N/m}$ is in a relaxed state initially. Now the lower piston is moved upwards by $h/2$ where h is the initial length of gas column. It is observed that the upper piston moves up by a distance $h/16$. Find h and the final temperature of the gas. (Take $\gamma = 1.5$)



Solution Let P_0 be the external pressure. So, $P_0 = 10^5 \text{ Pa}$.

Since, the system initially is in equilibrium, the initial pressure of the gas is

$$P_i = P_0 = 10^5 \text{ Pa}$$



The initial and final volumes of the gas are respectively

$$V_i = Ah \text{ and } V_f = A \left(h - \frac{h}{2} + \frac{h}{16} \right) = \frac{9}{16} Ah$$

As the gas undergoes adiabatic process,

$$P_i V_i^\gamma = P_f V_f^\gamma$$

$$\Rightarrow P_f = \left(\frac{V_i}{V_f} \right)^\gamma P_i = \left(\frac{16}{9} \right)^{1.5} \times 10^5 = \frac{64}{27} \times 10^5 \text{ Pa}$$

As the upper piston is in equilibrium,

$$\frac{kh}{16} = (P_f - P_0) A = \left(\frac{64}{27} \times 10^5 - 10^5 \right) \times 27 \times 10^{-4} = 370$$

$$\therefore h = \frac{370 \times 16}{3700} = \boxed{1.6 \text{ m}}$$

From $\frac{P_i V_i}{T_i} = \frac{P_f V_f}{T_f}$, we have

$$T_f = \left(\frac{P_f}{P_i} \right) \left(\frac{V_i}{V_f} \right) T_i = \frac{64}{27} \times \frac{9}{16} \times 273 = \boxed{364 \text{ K}}$$

9. The figure shows an insulated cylinder divided into three parts, A, B and C. Pistons I and II are connected by a rigid rod and can move without friction inside the cylinder. Piston I is perfectly conducting while piston II is perfectly insulating. The initial state of the gas ($\gamma = 1.5$) present in each compartment A, B and C is as shown. Now, compartment A is slowly given heat through a heater H such that the final volume of C becomes $4V_0/9$. Assuming the gas to be ideal, find

- final pressures and final temperatures in each compartment A, B and C
- heat supplied by the heater
- work done by gas in A and B and heat flowing across piston

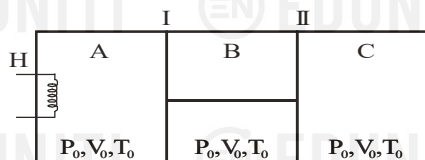


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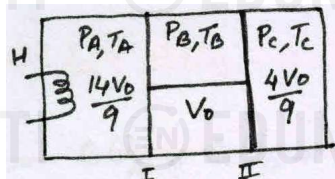
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Solution. The final situation is shown in the figure.

(a) As the total volume and the volume of B remains same,

$$V_A = 2V_0 - V_C = 2V_0 - 4V_0/9 = \boxed{14V_0/9}$$



The process in compartment C is adiabatic.

$$\text{So, } P_0 V_0^\gamma = P_C (4V_0/9)^\gamma$$

$$\therefore P_C = P_0 (9/4)^{1.5} = \boxed{27P_0/8}$$

$$\text{Also, } T_0 V_0^{\gamma-1} = T_C (4V_0/9)^{\gamma-1}$$

$$\therefore T_C = T_0 (9/4)^{1.5-1} = \boxed{3T_0/2}$$

As the system is finally in equilibrium,

$$P_A = P_C = 27P_0/8$$

$$\text{Since, } \frac{P_0 V_0}{T_0} = \frac{P_A V_A}{T_A} = \frac{(27P_0/8)(14V_0/9)}{T_A}, \text{ we have}$$

$$T_A = \frac{27}{8} \times \frac{14}{9} T_0 = \boxed{\frac{21}{4} T_0}$$

$$\text{As piston I is conducting, } T_B = T_A = 21T_0/4$$

$$\text{Since, } \frac{P_0 V_0}{T_0} = \frac{P_B V_B}{T_B} = \frac{P_B V_0}{21T_0/4}, \text{ we have } P_B = \boxed{\frac{21}{4} P_0}$$

(b) Since, the total volume of the entire system is constant, $\Delta W = 0$

$$\Rightarrow \Delta Q = \Delta U = \Delta U_A + \Delta U_B + \Delta U_C$$

$$\Rightarrow nC_V [\Delta T_A + \Delta T_B + \Delta T_C]$$

$$= \frac{nR}{\gamma-1} \left[\left(\frac{21}{4} T_0 - T_0 \right) + \left(\frac{21}{4} T_0 - T_0 \right) + \left(\frac{3}{2} T_0 - T_0 \right) \right]$$

$$= \frac{nR}{1.5-1} \left[\frac{17}{4} T_0 + \frac{17}{4} T_0 + \frac{T_0}{2} \right] = 18nRT_0$$

$$\therefore \boxed{\Delta Q = 18P_0 V_0}$$

(c) As the volume of gas B is constant, work done by gas in B,

$$\boxed{\Delta W_B = 0}$$

$$\Rightarrow \Delta Q_B = \Delta U_B + \Delta W_B = nC_V \Delta T_B + 0$$

$$= \frac{nR}{1.5-1} \left(\frac{21}{4} T_0 - T_0 \right) = \frac{17}{2} nRT_0 = \frac{17}{2} P_0 V_0$$

$$\therefore \text{Heat flowing across piston I is } \Delta Q_B = \boxed{\frac{17}{2} P_0 V_0}$$

$$\Delta Q_A = \Delta Q - \Delta Q_B = 18P_0 V_0 - \frac{17}{2} P_0 V_0 = \frac{19}{2} P_0 V_0$$

$$\Delta U_A = nC_V \Delta T_A = \frac{nR}{1.5-1} \left(\frac{21}{4} T_0 - T_0 \right) = \frac{17}{2} P_0 V_0$$

$\therefore$ Work done by gas in A,

$$\Delta W_A = \Delta Q_A - \Delta U_A = \frac{19}{2} P_0 V_0 - \frac{17}{2} P_0 V_0 = \boxed{P_0 V_0}$$

Paragraph Type

Passage for Question Nons. 10 to 12

An ideal diatomic gas is expanded so that the amount of heat transferred to the gas is equal to the decrease in its internal energy.

10. The molar specific heat of the gas in this process is given by C whose value is

- (a) $-5R/2$ (b) $-3R/2$
(c) $2R$ (d) $5R/2$

11. The process can be represented by the equation $TV^n = \text{constant}$, where the value of n is

- (a) $7/5$ (b) $1/5$
(c) $3/2$ (d) $3/5$

12. If in the above process, the initial temperature of the gas is T_0 and the final volume is 32 times the initial volume, the work done (in joules) by one mole of gas during the process will be

- (a) $\frac{RT_0}{2}$ (b) $\frac{5RT_0}{2}$
(c) $2RT_0$ (d) RT_0

Passage Solution

10(a) It is given that $\Delta Q = -\Delta U$

$$\Rightarrow nC \Delta T = -nC_V \Delta T \quad \therefore C = -C_V = -5R/2$$

11(b) For a polytropic process, $PV^\alpha = \text{const}$ or $TV^{\alpha-1} = \text{const}$,

$$C = C_V + \frac{R}{1-\alpha} \Rightarrow -\frac{5}{2}R = \frac{5}{2}R + \frac{R}{1-\alpha}$$

$$\Rightarrow \alpha = 6/5 \quad \therefore n = \alpha - 1 = 1/5$$

12(b) For this process, $T_1 V_1^{1/5} = T_2 V_2^{1/5}$

$$\Rightarrow T_2 = (V_1/V_2)^{1/5} T_1 = (1/32)^{1/5} T_0 = T_0/2$$

$$\therefore \Delta W = \frac{nR(T_2 - T_1)}{1-\alpha} = \frac{1 \times R(T_0/2 - T_0)}{1-6/5} = \frac{5}{2} RT_0$$

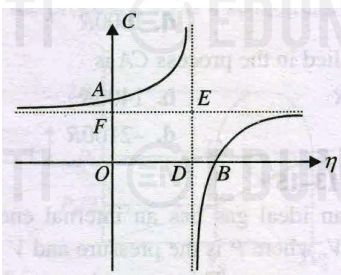
Passage for Question Nons. 13 to 15

A process in which work performed by an ideal gas is proportional to the corresponding change in its internal energy is described as a polytropic process. If we represent work done by a polytropic process by ΔW and increase in internal energy as ΔU , then $\Delta W \propto \Delta U$

For this process, it can be shown that the relation between pressure and volume is given by the equation

$$PV^\alpha = \text{constant}$$

In polytropic process, the variation of molar specific heat C with α for a monatomic gas is plotted as shown in the graph.



13 In the graph, the y-coordinate of point A is

- (a) $3R/2$ (b) $5R/2$
(c) $7R/2$ (d) $4R$

14. In the graph, the x-coordinate of point B is

- (a) $7/5$ (b) $5/3$
(c) $2/3$ (d) $8/3$

15. For a monatomic gas, the values of polytropic constant α for which molar specific heat is negative is

- (a) $1 < \alpha < 2/3$ (b) $1 < \alpha < 8/3$
(c) $1 < \alpha < 5/3$ (d) $2/3 < \alpha < 8/3$

Passage Solution

13(b) For polytropic process, $PV^\alpha = \text{const}$, molar specific heat,

$$C = C_v + \frac{R}{1-\alpha} = \frac{3}{2}R + \frac{R}{1-\alpha}$$

At A, $\alpha = 0$. $\therefore C = 3R/2 + R = 5R/2$

14(b) At B, $C = 0 \Rightarrow 0 = \frac{3R}{2} + \frac{R}{1-\alpha} \therefore \alpha = \frac{5}{3}$

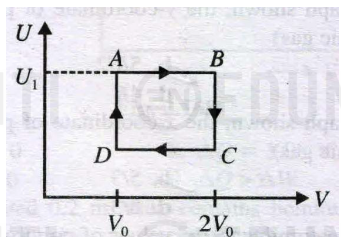
15(c) $C < 0$

$$\Rightarrow \frac{3R}{2} - \frac{R}{\alpha-1} < 0 \Rightarrow \frac{3\alpha-3-2}{2(\alpha-1)} < 0$$

$$\Rightarrow \frac{3\alpha-5}{\alpha-1} < 0 \therefore 1 < \alpha < \frac{5}{3}$$

Passage for Question Nons. 16 to 18

One mole of an ideal gas has an internal energy given by $U = U_0 + 2PV$, where P is the pressure and V is the volume of the gas. U_0 is a constant. This gas undergoes the quasi-static cyclic process ABCD as shown in the $U-V$ diagram.



16. The molar heat capacity of the gas at constant pressure is

- (a) $2R$ (b) $3R$
(c) $\frac{5}{2}R$ (d) $4R$

17. The work done by the ideal gas in the process AB is

- (a) zero (b) $U_1 - U_0$
(c) $\frac{U_1 - U_0}{2}$ (d) $\left(\frac{U_1 - U_0}{2}\right) \ln 2$

18. The gas must be

- (a) monatomic
(b) diatomic
(c) a mixture of mono and diatomic gases
(d) a mixture of di- and tri-atomic gases

Passage Solution

16(b) $U = U_0 + 2PV = U_0 + 2nRT$

$$\Rightarrow \Delta U = 2nR\Delta T = nC_p\Delta T \Rightarrow C_p = 2R$$

$$\therefore C_p = C_v + R = 3R$$

17(d) For process AB, $U = \text{const.} \Rightarrow T = \text{const.}$

$$\therefore \Delta W_{AB} = nRT \ln \frac{2V_0}{V_0} = PV \ln 2 = \left(\frac{U_1 - U_0}{2}\right) \ln 2$$

18(c) As $C_p = 2R$, out of the given options, the gas must be a mixture of monatomic ($C_v = 1.5R$) and diatomic ($C_v = 2.5R$) gases.

Passage for Question Nons. 19 to 20

A fluid system undergoes a process following $p = \frac{5}{V} + 1.5$, where p is in bar and V is in m^3 . During the process, volume changes from 0.15 m^3 to 0.05 m^3 and the system rejects 45 kJ of heat.

19. Work done during the process is (Take, $\log_e 3 = 1.098$)

- (a) -564 kJ (b) 564 kJ
(c) 483 kJ (d) -483 kJ

20. Change in internal energy during the process is

- (a) -45 kJ (b) 45 kJ
(c) 519 kJ (d) -519 kJ

(Passage with Solution)

19(a) $P = \left(\frac{5}{V} + 1.5\right) \text{ bars} = \left(\frac{5}{V} + 1.5\right) \times 10^5 \text{ pascals}$

$$\Delta W = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \left(\frac{5}{V} + 1.5\right) \times 10^5 dV$$

Here, $V_1 = 0.15 \text{ m}^3$ and $V_2 = 0.05 \text{ m}^3$

$$\therefore \Delta W = [-5 \ln 3 + 1.5(0.05 - 0.15)] \times 10^5$$

$$= (-5 \times 1.098 - 0.15) \times 10^5 = -5.64 \times 10^5 \text{ J}$$

$$= -564 \text{ kJ}$$

20(c) From 1st law of thermodynamics,

$$\Delta U = \Delta Q - \Delta W = (-45) - (-564) = 519 \text{ kJ}$$



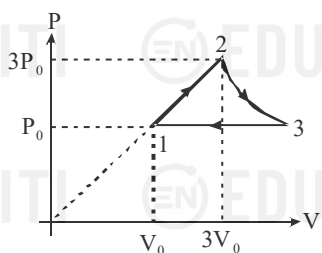
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Passage for Question Nons. 21 to 23

One mole of an ideal monoatomic gas undergoes a cyclic process as shown in figure. Temperature at point 1 is 300 K and process $2 \rightarrow 3$ is isothermal.



21. Net work done by gas in complete cycle is

- (a) $(9 \ln 3 + 12) P_0 V_0$ (b) $(9 \ln 3 + 4) P_0 V_0$
(c) $(9 \ln 3 - 4) P_0 V_0$ (d) $(9 \ln 3 - 8) P_0 V_0$

22. Heat capacity of process $1 \rightarrow 2$ is

- (a) $\frac{R}{2}$ (b) $\frac{3R}{2}$
(c) $\frac{5R}{2}$ (d) $2R$

23. The efficiency of cycle is

- (a) $\left(\frac{9 \ln 3 + 4}{9 \ln 3 + 12} \right)$ (b) $\left(\frac{9 \ln 3 - 4}{9 \ln 3 + 12} \right)$
(c) $\left(\frac{9 \ln 3 - 4}{9 \ln 3 + 16} \right)$ (d) $\left(\frac{9 \ln 3 + 12}{9 \ln 3 + 16} \right)$

Passage Solution

21(c) As process $2 \rightarrow 3$ is isothermal,

$$P_2 V_2 = P_3 V_3 \Rightarrow (3P_0)(3V_0) = P_0 V_3$$

$$\Rightarrow V_3 = 9V_0$$

$$\Delta W = \Delta W_{12} + \Delta W_{23} + \Delta W_{31}$$

$$= \frac{1}{2}(3P_0 + P_0)(3V_0 - V_0) + nRT \ln 3 - P_0(9V_0 - V_0)$$

$$= 4P_0 V_0 + (3P_0)(3V_0) \ln 3 - 8P_0 V_0 = P_0 V_0 (9 \ln 3 - 4)$$

22(d) Let the temperature at 1 be T_0 . Now

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \Rightarrow \frac{P_0 V_0}{T_0} = \frac{(3P_0)(3V_0)}{T_2} \Rightarrow T_2 = 9T_0$$

For process $1 \rightarrow 2$,

$$\Delta Q = nC_V \Delta T + \Delta W + n \left(\frac{3R}{2} \right) (9T_0 - T_0) + 4P_0 V_0$$

$$= 12nRT_0 + 4nRT_0 = 16nRT_0$$

$$\therefore C = \frac{\Delta Q}{n \Delta T} = \frac{16nRT_0}{n(9T_0 - T_0)} = 2R$$

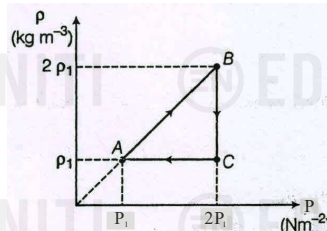
23(c) $\Delta Q_{in} = \Delta Q_{12} + \Delta Q_{23} = 16nRT_0 + \Delta W_{23}$

$$= 16P_0 V_0 + 9P_0 V_0 \ln 3 = P_0 V_0 (9 \ln 3 + 16)$$

$$\therefore \eta = \frac{\Delta W_{23}}{\Delta Q_{in}} = \frac{P_0 V_0 (9 \ln 3 - 4)}{P_0 V_0 (9 \ln 3 + 16)} = \frac{9 \ln 3 - 4}{9 \ln 3 + 16}$$

Passage for Question Nons. 24 to 25

Consider a thermodynamic cycle as shown in which a monoatomic ideal gas is taken from state A to B to C to A.



24. Ratios of work done in the processes, AB, BC, and CA

($W_{AB} : W_{BC} : W_{CA}$) are

- (a) $\ln 2 : 1 : 1$ (b) $-\ln 2 : 1 : 0$
(c) $-\ln 2 : 1 : 0$ (d) $-\ln 2 : 1 : 1$

25. Ratio of heat rejected by gas to the total heat supplied is

- (a) $\frac{2}{5}(1 - \ln 2)$ (b) $\frac{5}{2}(1 - \ln 2)$
(c) $\frac{2}{5} \ln 2$ (d) $\left(\frac{3 + \ln 4}{5} \right)$

Passage Solution

24(b) In process AB, $P \propto 1/\rho \propto m/V$. So, process is isothermal

Process BC is isobaric.

In process CA, $\rho = m/V = \text{const.}$ So process is isochoric.

Let $T_A = T_0$. Then $T_B = T_0$ and $T_C = 2T_0$

$$\Rightarrow W_{AB} = nRT_0 \ln(P_A/P_B) = -nRT_0 \ln 2,$$

$$W_{BC} = nR(2T_0 - T_0) = nRT_0 \text{ and } W_{CA} = 0$$

$$\therefore W_{AB} : W_{BC} : W_{CA} = -\ln 2 : 1 : 0$$

25(d) Total heat rejected = $|\Delta Q_{AB}| + |\Delta Q_{CA}|$

$$= |\Delta W_{AB}| + |nC_V(T_0 - 2T_0)|$$

$$= nRT_0 \ln 2 + \frac{3}{2} nRT_0$$

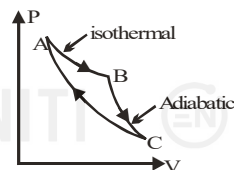
$$\text{Heat supplied} = \Delta Q_{BC} = nC_P(2T_0 - T_0) = \frac{5}{2} nRT_0$$

$$\therefore \frac{\text{Heat rejected}}{\text{Heat supplied}} = \frac{nRT_0(\ln 2 + 3/2)}{nRT_0(5/2)} = \frac{3 + \ln 4}{5}$$

Passage for Question Nons. 26 to 27

A cyclic process for an ideal gas is shown in figure. Given

$$\Delta W_{AB} = +700 \text{ J}, \Delta W_{BC} = +400 \text{ J}, \Delta Q_{CA} = -100 \text{ J}.$$



26. The work done in process CA is

- (a) -500 J (b) 500 J
(c) 400 J (d) -400 J

27. The efficiency of the cycle is

- (a) 100% (b) 83.44%
(c) 85.71% (d) 81.11%

Passage Solution



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26(a) $\Delta W_{AB} = 700 \text{ J}$, $\Delta W_{BC} = 400 \text{ J}$ and $\Delta Q_{CA} = -100 \text{ J}$

Since, AB is isothermal, $\Delta Q_{AB} = \Delta W_{AB} = 700 \text{ J}$

Since, BC is adiabatic, $\Delta Q_{BC} = 0$

For complete cycle, $\Delta U = 0 \Rightarrow \Delta Q = \Delta W$

$$\Rightarrow \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA} = \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CA}$$

$$\Rightarrow 700 + 0 - 100 = 700 + 400 + \Delta W_{CA}$$

$$\therefore \Delta W_{CA} = -500 \text{ J}$$

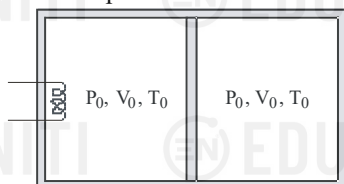
27(c) $\Delta W = \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CA} = 700 + 400 - 500 = 600 \text{ J}$

$$\Delta Q_{in} = \Delta Q_{AB} = 700 \text{ J}$$

$$\therefore \eta = \frac{\Delta W}{\Delta Q_{in}} = \frac{600}{700} = 0.8571 = 85.71\%$$

Passage for Question Nons. 28 to 30

One mole of a monoatomic ideal gas occupies two chambers of a cylinder partitioned by means of a movable and frictionless piston. The walls of the cylinder as well as the piston are thermal insulators. Initially equal amounts of gas fill both the chambers at (P_0, V_0, T_0) . The left chamber is slowly heated by an electric heater. The gas in the left chamber expands, pushing the piston to the right. The gas on the right chamber is compressed until the pressure becomes $32 P_0$.



28. The final volume of left chamber is

- (a) $\frac{V_0}{8}$ (b) $\frac{15}{8} V_0$
(c) $\frac{7}{8} V_0$ (d) $\frac{9}{8} V_0$

29. The work done on the gas in the right chamber is

- (a) $\frac{9}{2} P_0 V_0$ (b) $-\frac{9}{2} P_0 V_0$
(c) $\frac{13}{2} P_0 V_0$ (d) $\frac{17}{2} P_0 V_0$

30. The change in internal energy of the gas in the left chamber is

- (a) $\frac{131}{2} P_0 V_0$ (b) $\frac{177}{2} P_0 V_0$
(c) $59 P_0 V_0$ (d) $93 P_0 V_0$

Passage Solution

28. (b) The compression in the right chamber is adiabatic.

If V_R is the final volume of the gas in right chamber, then

$$P_0 V_0^{5/3} = 32 P_0 V_R^{5/3}$$

$$\Rightarrow V_R = \frac{V_0}{8} \quad \therefore V_L = 2V_0 - V_R = \frac{15}{8} V_0$$

29(a) Work done on the gas

$$= \frac{P_R V_R - P_0 V_0}{\gamma - 1} = \frac{32 P_0 (V_0/8) - P_0 V_0}{5/3 - 1} = \frac{9}{2} P_0 V_0$$

30(b) For mechanical equilibrium, $P_L = P_R = 32 P_0$

The change in the internal energy of the gas in the left chamber is

$$\begin{aligned} \Delta U &= n C_V \Delta T = n \times \frac{3}{2} R \Delta T = \frac{3}{2} \Delta(nRT) = \frac{3}{2} \Delta(PV) \\ &= \frac{3}{2} \Delta(P_L V_L - P_0 V_0) = \frac{3}{2} \left[(32 P_0) \left(\frac{15}{8} V_0 \right) - P_0 V_0 \right] \\ &= \frac{177}{2} P_0 V_0 \end{aligned}$$

More than one Correct

31. C_p is always greater than C_v due to the fact that
(a) no work is done on heating the gas at constant volume.
(b) when a gas absorbs heat at constant pressure, its volume must change so as to do some external work.
(c) for the same rise of temperature, the internal energy of a gas changes by equal amount at constant volume and at constant pressure.
(d) for the same rise of temperature, the internal energy of a gas changes by a smaller amount at constant volume than at constant pressure.
- 31.(a,b,c) For the same rise of temperature, internal energy of a gas changes by the same amount at constant volume and a constant pressure. However, at constant volume, no work is being done whereas at constant pressure, the gas absorbs extra heat to do some external work.

32 An ideal gas can be expanded from an initial state to a certain volume through two different processes

- (i) $PV^2 = \text{constant}$ and
(ii) $P = KV^2$ where K is a positive constant.

Then,

- (a) final temperature in (i) will be greater than in (ii)
(b) final temperature in (ii) will be greater than in (i)
(c) total heat given to the gas in (i) case is greater than in (ii)
(d) total heat given to the gas in (ii) case is greater than in (i)

32(b,d) For process (i), $PV^2 = \left(\frac{nRT}{V} \right) V^2 = nRTV = \text{const.}$

$$\Rightarrow T \propto 1/V$$

Similarly for process (ii), $KV^2 = P = \frac{nRT}{V}$

$$\Rightarrow T \propto V^3$$

During expansion, T decreases in case (i) and increases in case (ii). Hence, final temperature is greater in case (ii).

In case (i), $\Delta T < 0$. So heat is released by the gas.

In case (ii), $\Delta T > 0$. So heat is absorbed by the gas.

So, the total heat given to the gas is greater in case (ii).

33 n moles of an ideal triatomic linear gas undergoes a process in which the temperature changes with volume as $T = k_1 V^2$ where k_1 is a constant. Choose the correct alternatives

- (a) At normal temperature $C_v = \frac{5}{2}R$
- (b) At any temperature $C_p - C_v = R$
- (c) At normal temperature molar heat capacity $C = 3R$
- (d) At any temperature molar heat capacity $C = 3R$

33(a, b, c) Degree of freedom, $f = 5$

$$\text{So, at normal temperature, } C_v = \frac{f}{2}R = \frac{5}{2}R$$

$$C_p - C_v = R \text{ is for any temperature.}$$

$$T = k_1 V^2 \Rightarrow \frac{PV}{nR} = k_1 V^2$$

$$\Rightarrow PV^{-1} = \text{const}$$

This is a polytropic process where $\alpha = -1$. Its molar heat capacity at normal temperature is

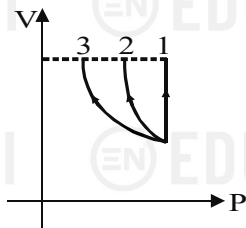
$$C = C_v + \frac{R}{1-\alpha} = \frac{5}{2}R + \frac{R}{1+1} = 3R$$

At higher temperatures, vibrational energies also come into play. Thus degree of freedom, f increases. So, option (d) is incorrect

34 The figure shows V-P curve for three processes.

Choose the correct statement(s)

- (a) Work done is maximum in process 1.

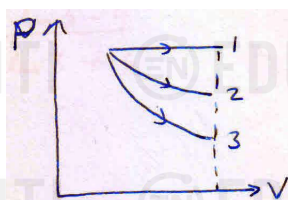


- (b) Temperature must increase in process 2 and 3.
- (c) Heat must be supplied in process 1.
- (d) If final volume of gas in process 1, 2 and 3 are same, then temperature must be same.

34(a, c) The curve is redrawn as shown in figure. Area under P-V curve and Volume axis represents work. It is maximum in process 1.

Process 1 is isobaric. So, $\Delta Q > 0$.

Processes 2 and 3 are of the form $PV^\alpha = \text{const}$.



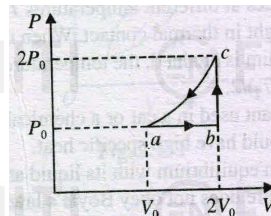
If $\alpha = 1$, process is isothermal and $\Delta T = 0$.

If $\alpha > 1$, $\Delta T > 0$ and if $\alpha < 1$, $\Delta T < 0$.

If $\alpha = \gamma$, process is adiabatic and $\Delta Q = 0$

If $\alpha < \gamma$, $\Delta Q > 0$ and if $\alpha > \gamma$, $\Delta Q < 0$

35 One mole of an ideal monatomic gas having initial temperature T_0 at a, is made to go through the cycle abca shown in fig. If U denotes the internal energy, then choose the correct alternative/s



$$(a) U_c > U_b > U_a \quad (b) U_c - U_b = 3RT_0$$

$$(c) U_c - U_a = \frac{9RT_0}{2} \quad (d) U_b - U_a = \frac{3RT_0}{2}$$

35(a, b, c, d) $a \rightarrow b$ is an isobaric process where

$$V \propto T \Rightarrow T_b = 2T_a = 2T_0$$

$b \rightarrow c$ is an isochoric process where

$$P \propto T \Rightarrow T_c = 2T_b = 4T_0$$

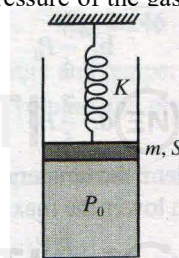
$$U = nC_v T = \frac{3}{2}nRT = \frac{3}{2}RT$$

$$\Rightarrow U_a = \frac{3}{2}RT_0, U_b = 3RT_0 \text{ and } U_c = 6RT_0$$

$$\therefore U_c > U_b > U_a, U_c - U_b = 3RT_0, U_c - U_a = \frac{9}{2}RT_0$$

$$\text{and } U_b - U_a = \frac{3}{2}RT_0$$

36 In the arrangement shown in fig., gas is thermally insulated. An ideal gas is filled in the cylinder having pressure P_0 ($>$ atmospheric pressure P_a). The spring of force constant K is initially unstretched. The piston of mass m and area A is frictionless. In equilibrium, the piston rises up by distance x_0 , its speed is v and the pressure of the gas is P . Then,



$$(a) P = P_a + \frac{Kx_0}{A} + \frac{mg}{A}$$

(b) work done by the gas from initial to equilibrium position is

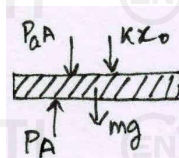
$$\frac{1}{2}Kx_0^2 + mgx_0 + \frac{1}{2}mv^2$$

(c) decrease in internal energy of the gas from initial to

$$\text{equilibrium position is } P_a Ax_0 + \frac{1}{2}Kx_0^2 + mgx_0 + \frac{1}{2}mv^2$$

(d) all of the above

36(a, c)





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In equilibrium, the net force on the piston must be zero.

$$\Rightarrow PA = P_a A + Kx_0 + mg \quad \therefore P = P_a + \frac{Kx_0}{A} + \frac{mg}{A}$$

As per work energy theorem, $W_{\text{gas}} + W_{\text{atm}} = \Delta U + \Delta K$

$$\Rightarrow W_{\text{gas}} - P_a Ax_0 = \frac{1}{2} Kx_0^2 + mgx_0 + \frac{1}{2} mv^2$$

$$\therefore W_{\text{gas}} = P_a Ax_0 + \frac{1}{2} Kx_0^2 + mgx_0 + \frac{1}{2} mv^2$$

As the process is adiabatic, $\Delta U = -\Delta W$

$\therefore$ Decrease in internal energy

$$= P_a Ax_0 + \frac{1}{2} Kx_0^2 + mgx_0 + \frac{1}{2} mv^2$$